PSA - Carry Forward

Agenda:

1. Count of AG pairs
2. Count of Subarrays
3. Print All Subarrays.
4. Sum of every Subarray

---

① Count of AG Pairs

```
long countA = 0;
long ans = 0;
for (int i = 0 ; i < s.length (); i++){
        char c = s.charAt(i);
        if (c == 'a'){
            countA++;   // carry forward the count
                           of a's
        }
        else if (c == 'g') {
            ans += countA;   // every previous 'a'
                                makes a pair
        }
    }
return ans;
```

---

② Print all Subarrays (Carry-forward the built subarray)

```
int n = A.length
for (int start=0; start<n ; start++){
        String corr = "";   //carry forward growing subarray
    for (int end = start ; end < n; end++){
        corr += A[end] + " ";
        Sop ln (corr
    }
}
```

③ Sum of every Subarray (carry-forward current sum)

```
int n = A.length
for(int start =0; start <n ; start++){
        int currSum = 0;
    for(int end=start ; end <n ; end++){
            currSum += A[end];      // carry-forward sum
            sop(currSum);
    }
}
```

Sum of all subarrays by this method will take $O(N^2)$

DSA - Contribution Techique and Sliding Window